

THE DATAGOOD(R) PACKAGE: AN OPEN SCIENCE APPROACH TO DATA ENGINEERING

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CUSTOMIZING LAYOUTS

Flexible Layouts

- The RG and DD QMD templates include a built-in style sheet that divides the page into sections
- Each section is populated by variable attributes (text descriptors for dictionary elements or data summaries)
- Each data type leverages different sections of the layout to organize content
- Positions of content and transformer functions are defined in the layout object
- The layout object will be embedded within the QMD template file so that users can adapt as needed
- Custom formatting functions can be defined by the user

layout <-

```
c( "div2 ;; vlabel    ;; LABEL          ;; v_to_txt",
    "div3 ;; vtype    ;; DATA TYPE    ;; v_to_txt",
    "div3 ;; scope    ;; SCOPE          ;; v_to_txt",
    "div4 ;; desc     ;; DESCRIPTION    ;; v_to_txt",
    "div4 ;; flevels  ;; LEVELS         ;; f_to_txt",
    "div4 ;; glevels  ;; ' '           ;; v_to_txt",
    "div4 ;; loc      ;; LOCATION CODE  ;; v_to_txt",
    "div5 ;; v        ;; STATS          ;; get_properties" )
```

DIV	VARIABLE	LABEL	FORMATTING FUNCTION
:----	:-----	:-----	:-----
div2	vlabel	LABEL	v_to_txt
div3	vtype	DATA TYPE	v_to_txt
div3	scope	SCOPE	v_to_txt
div4	desc	DESCRIPTION	v_to_txt
div4	flevels	LEVELS	f_to_txt
div4	glevels	' '	v_to_txt
div4	loc	LOCATION CODE	v_to_txt
div5	v	LABEL	get_properties

CSS Elements Controlling Layouts

```
.parent {  
  display: grid;  
  grid-template-columns: repeat(3, 1fr);  
  grid-template-rows: auto;  
  grid-column-gap: 20px;  
  grid-row-gap: 10px;  
  grid-template-areas:  
    "a a a"  
    "b b b"  
    "c d d"  
    "e f g"  
    "h h h"  
    "i i i"  
    "j k k"  
    "m o o"  
    "n o o"  
    "p p p";  
}
```

```
.div1 {  
  grid-area: a;  
  padding-bottom: 15px;  
}  
  
.div2 {  
  grid-area: b;  
  padding-left: 30px;  
}  
  
.div3 {  
  grid-area: c;  
  padding-left: 30px;  
}
```

Data Dictionary Format

- Accelerate the documentation and sharing of a dataset.
- The DGF provides a quick list-style template for providing the most important features of a dataset:
 - variable name
 - label
 - description / details
 - data types
 - factor level definitions
 - custom attributes (e.g. SCOPE)
- The data dictionary (DD) version limits documentation to dataset and variable definitions
- The research guide (RG) version includes data profiles

data dictionary format

VarName_o1_X

Ipsum is simply dummy text of the printing and typesetting industry.

DATA TYPE: **numeric**
SCOPE: **PZ**
LENGTH: **6**

DEFINITION: Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book.

VarName_o2_L2

Ipsum is simply dummy text of the printing and typesetting industry.

DATA TYPE: **factor**
SCOPE: **PC**
LENGTH: **24**

DEFINITION: Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book.

LEVELS:

A	goat
B	monkey
C	turtle
D	horse

VarName_o1_X

Ipsum is simply dummy text of the printing and typesetting industry.

DATA TYPE: numeric
SCOPE: PZ
LENGTH: 6

DEFINITION: Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book.

VarName_o2_L2

div1

div2 Ipsum is simply dummy text of the printing and typesetting industry.

DATA TYPE: factor
SCOPE: PC
LENGTH: 24

DEFINITION: Lorem Ipsum is simply dummy text of the printing and typesetting industry. Lorem Ipsum has been the industry's standard dummy text ever since the 1500s, when an unknown printer took a galley of type and scrambled it to make a type specimen book.

LEVELS:

A goat
B monkey
C turtle
D horse

div3

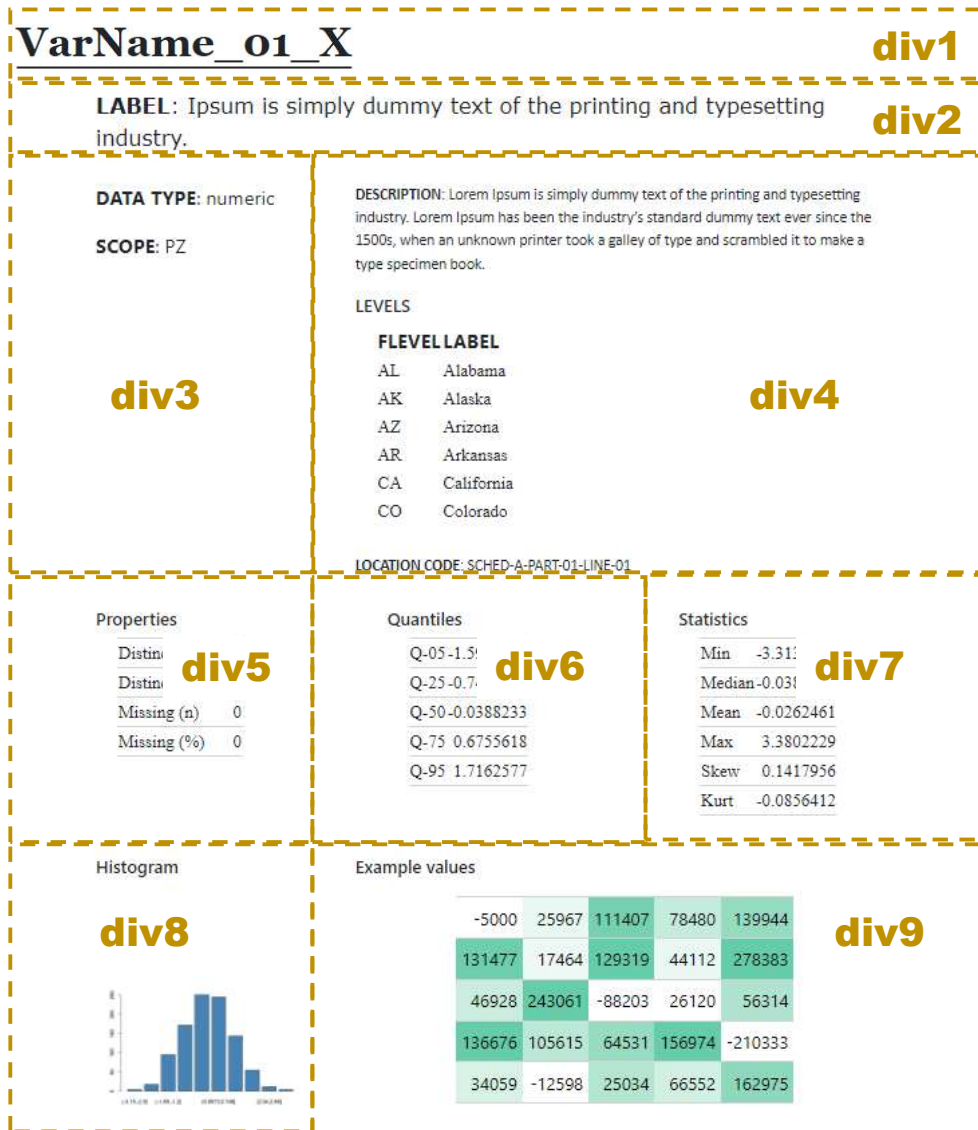
div4

data
dictionary
format

create_dd(dgf)
+
layout

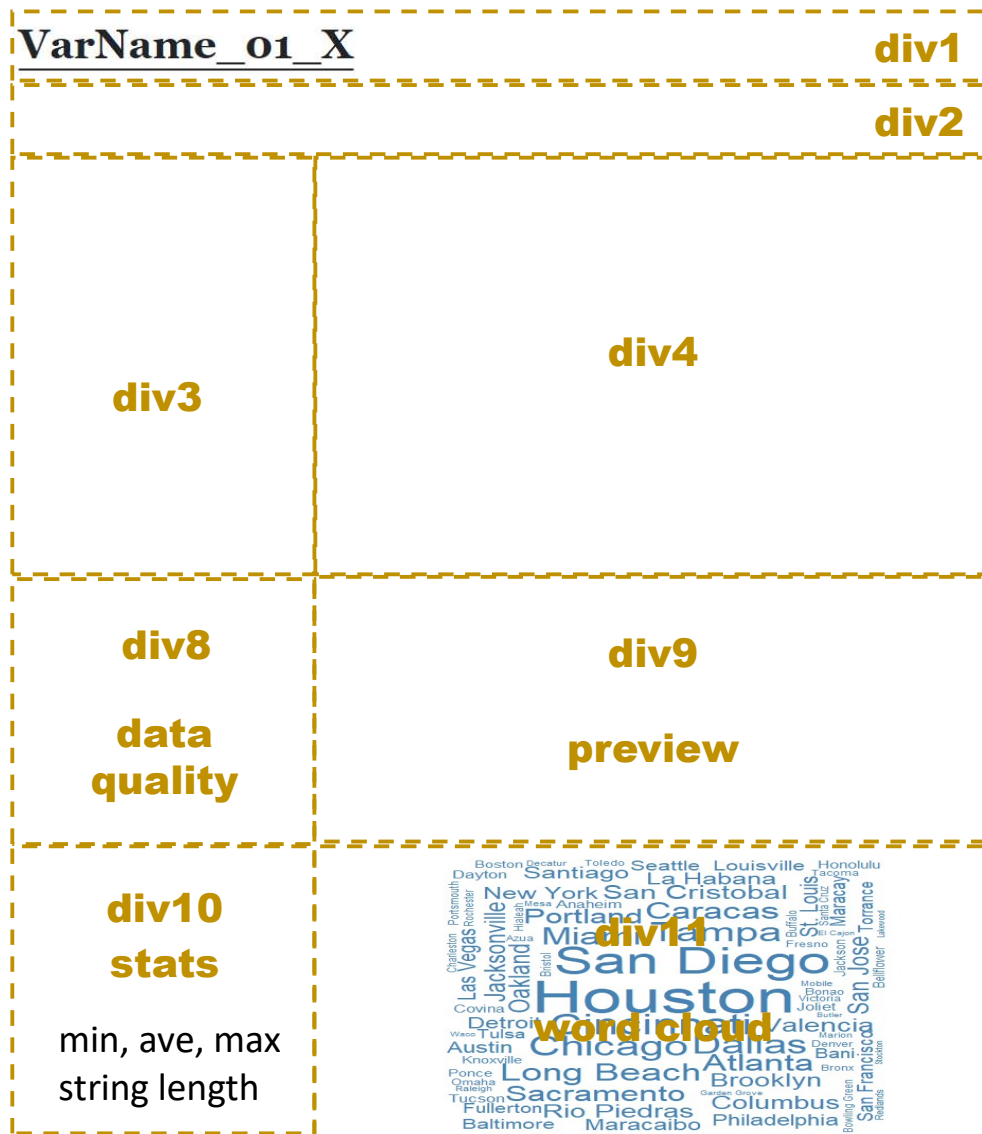
Research Guide Format

- The research guide provides more thorough documentation of a dataset than dictionary files
- The package manages data profiling functionality
- The QMD document can be turned into a true data user guide or public use file codebook that documents the data generation process and transformation
 - sampling framework, response rates, weights
 - survey questions or raw data fields
 - cleaning, standardization, normalization, and transformation processes applied
 - privacy protecting steps
 - etc

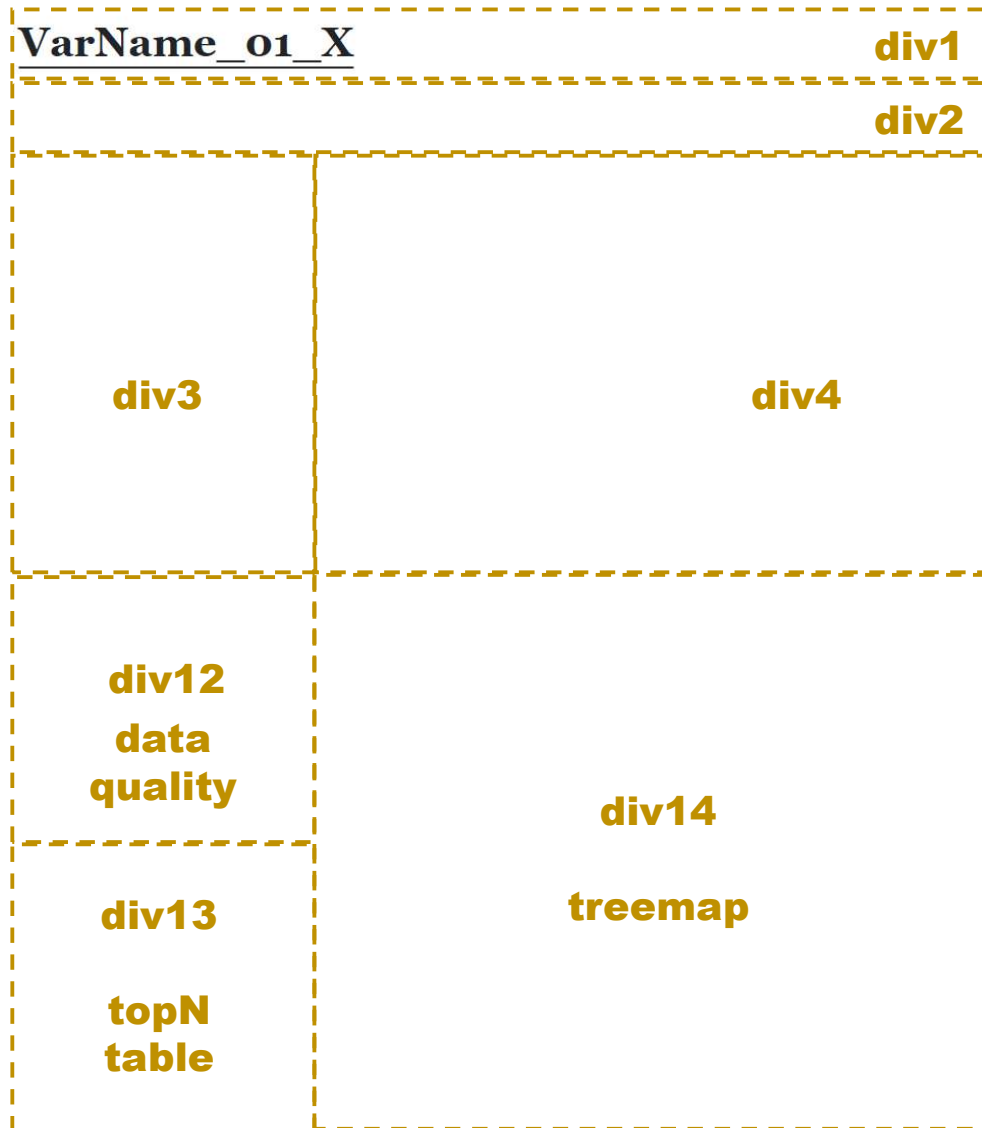


data profiling
layouts in the
RG.qmd file

data
profiling



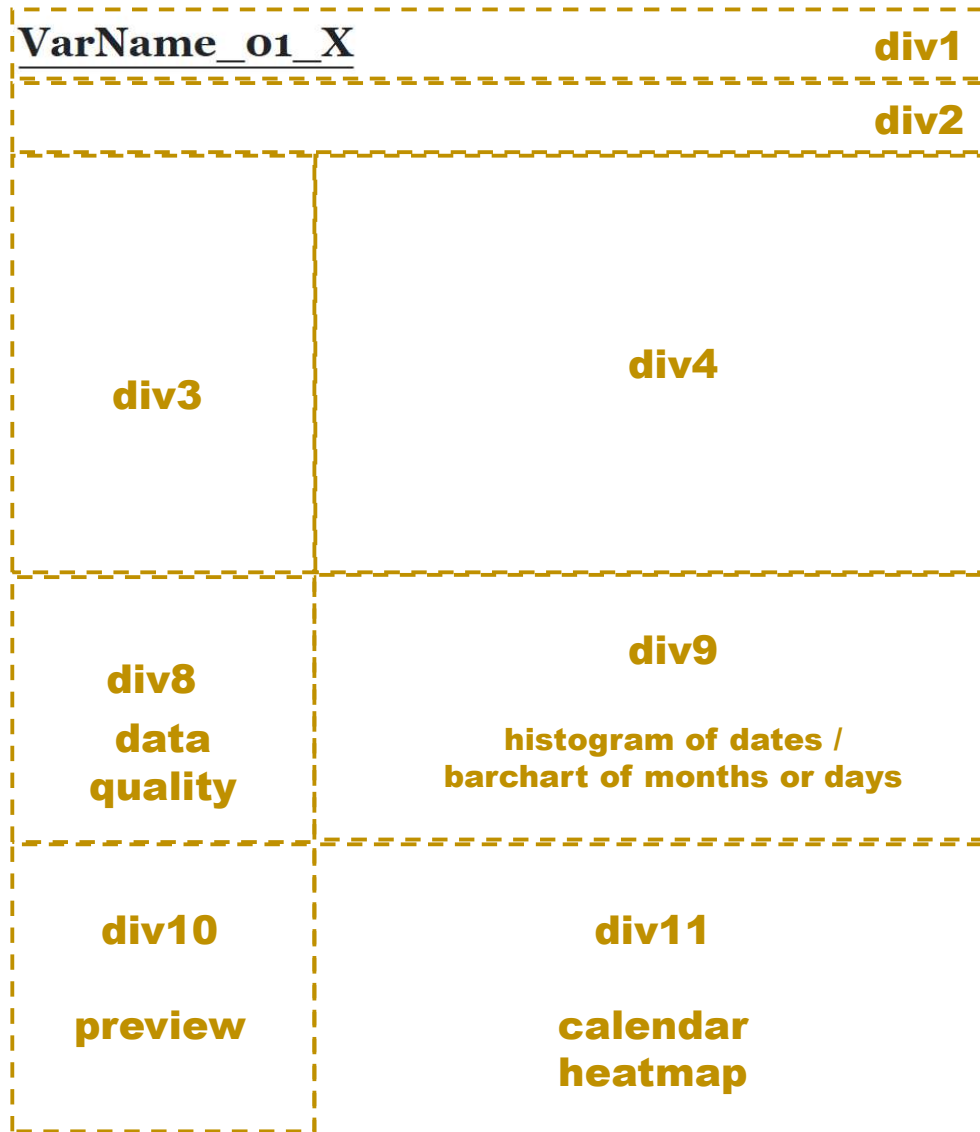
data type:
character



data type:
factor

<u>VarName_o1_X</u>			div1
			div2
div3		div4	
div5 data quality	div6 stats	div7 barchart	

data type:
logical (Boolean)



data type:
dates / time

ITERATIVE WORKFLOW

●

create_dgf(df.raw)

Create the DGF (user edits, adds labels and rules)

inspect_dgf(dgf)

Ensure json ok, rules are defined, etc.

ingest_raw(df.raw, dgf)

Apply the raw_convert rules, keep desired types.

data is
altered



standardize(df.v2, dgf)

update_dgf() Call after changes are made to data to ensure stored values are correct (e.g factor levels the same, update stored descriptive stats for numeric vars).

Apply standardization rules, update_dgf() called silently

data is
altered



create_vr(df.v3, dgf)

Create a data validation report using dgf_validate rules.

create_dd(df.v3, dgf)

Create a data dictionary from dgf fields.

create_rg(df.v3, dgf)

Create a research guide qmd, annotated before rendering.

format(df.v3, dgf)

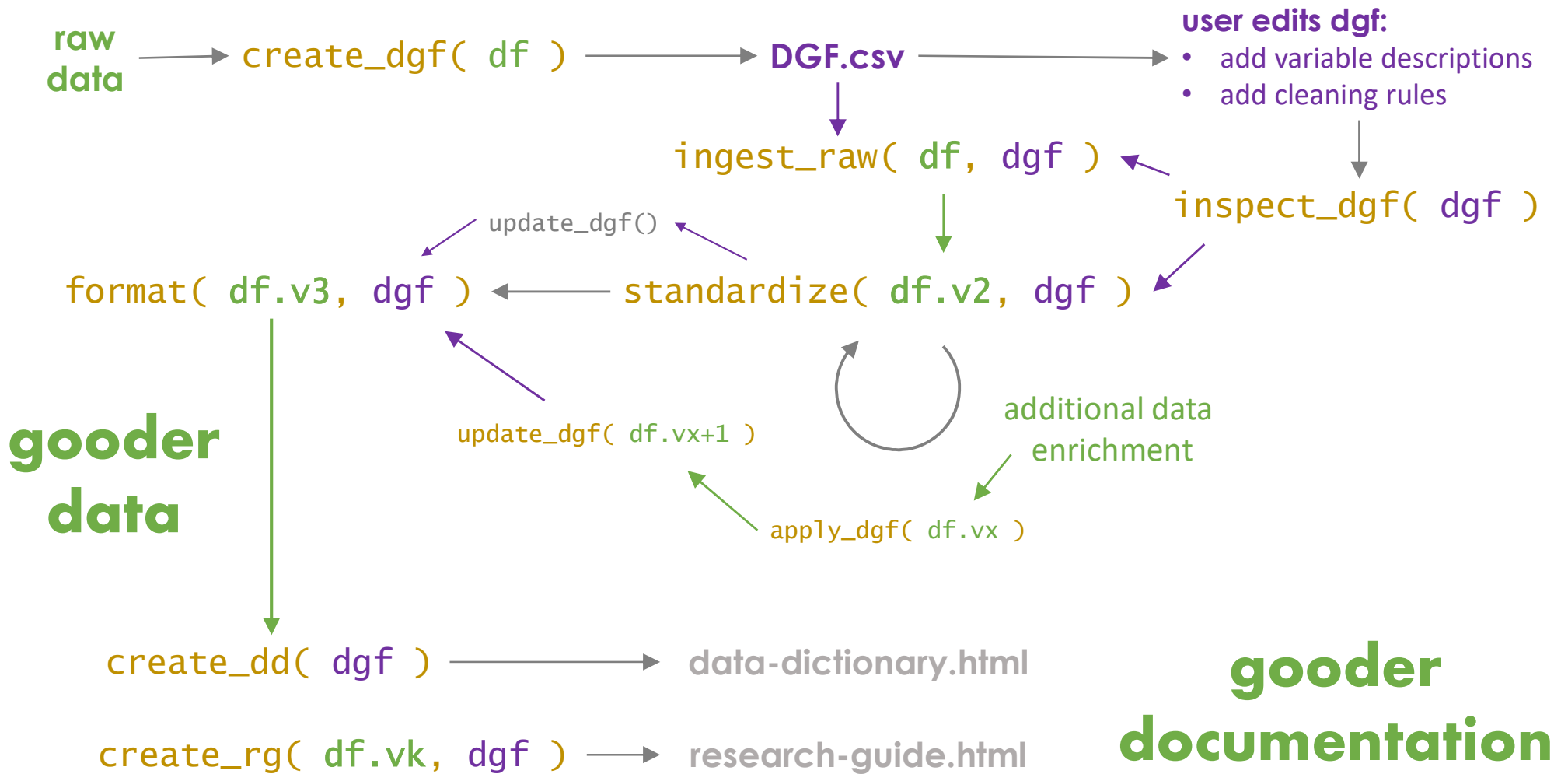
Apply formatting rules to make data pretty.



df.share

some context where the data is read, not analyzed, also used in dd & rg

can be
gener-
ated
any
time
once
DGF
exists



Default Graphical Summaries

Customized Graphics

Flexible Layouts

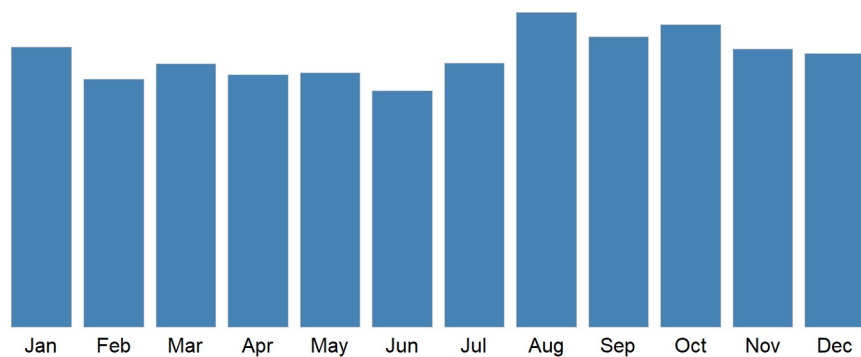
- Update the layout object by adding or substituting a section (div) label and function name
- Make the function available to the QMD template (not yet implemented)
 - Package will automatically try to source a DG.R script when rendering files. It will contain some of the default formatting and graphing functions for the user to customize, and they can add their own
 - Functions can be added directly to the QMD document and managed there

layout <-

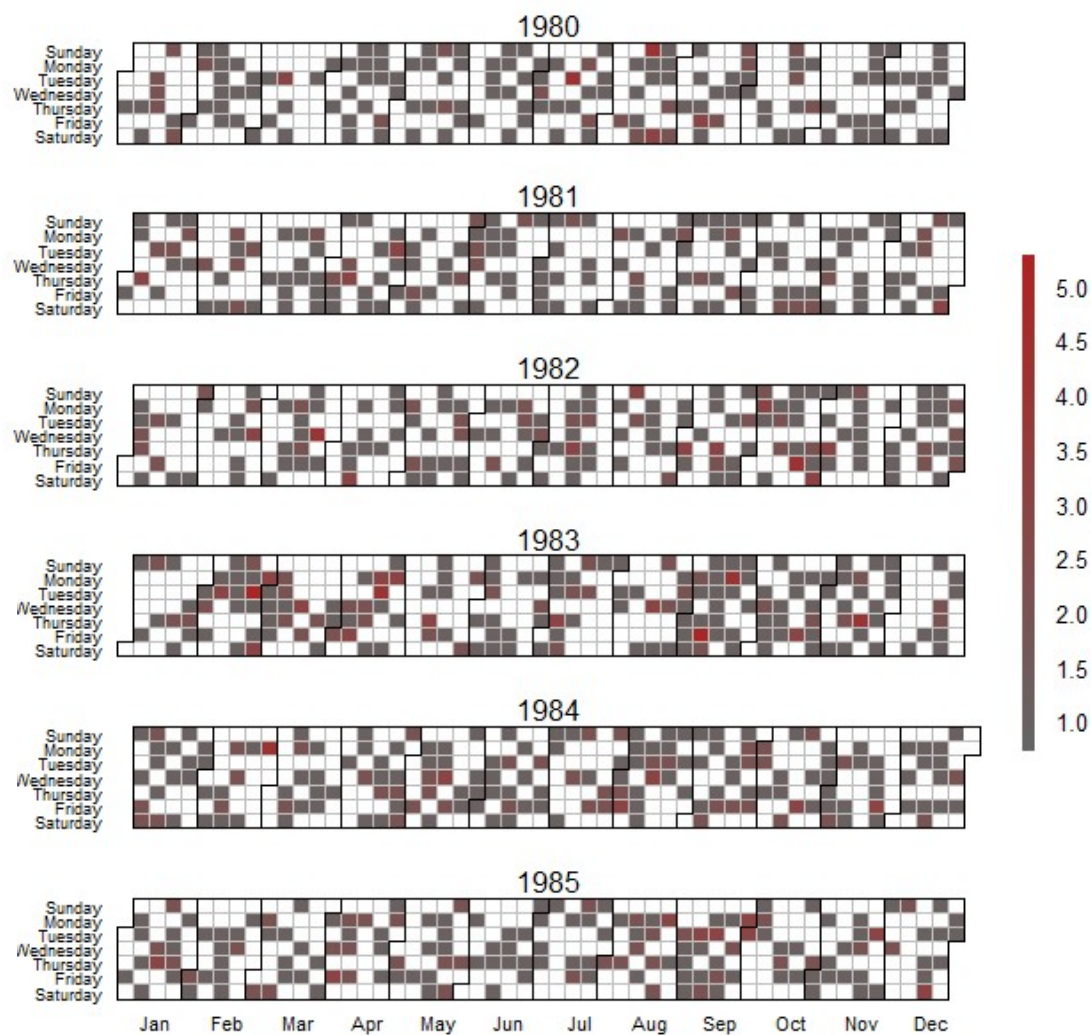
```
c( "div2 ;; vlabel    ;; LABEL          ;; v_to_txt",
    "div3 ;; vtype    ;; DATA TYPE     ;; v_to_txt",
    "div3 ;; scope    ;; SCOPE          ;; v_to_txt",
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div4	glevels	' '	v_to_txt
div4	loc	LOCATION CODE	v_to_txt
div5	v	LABEL	get_properties

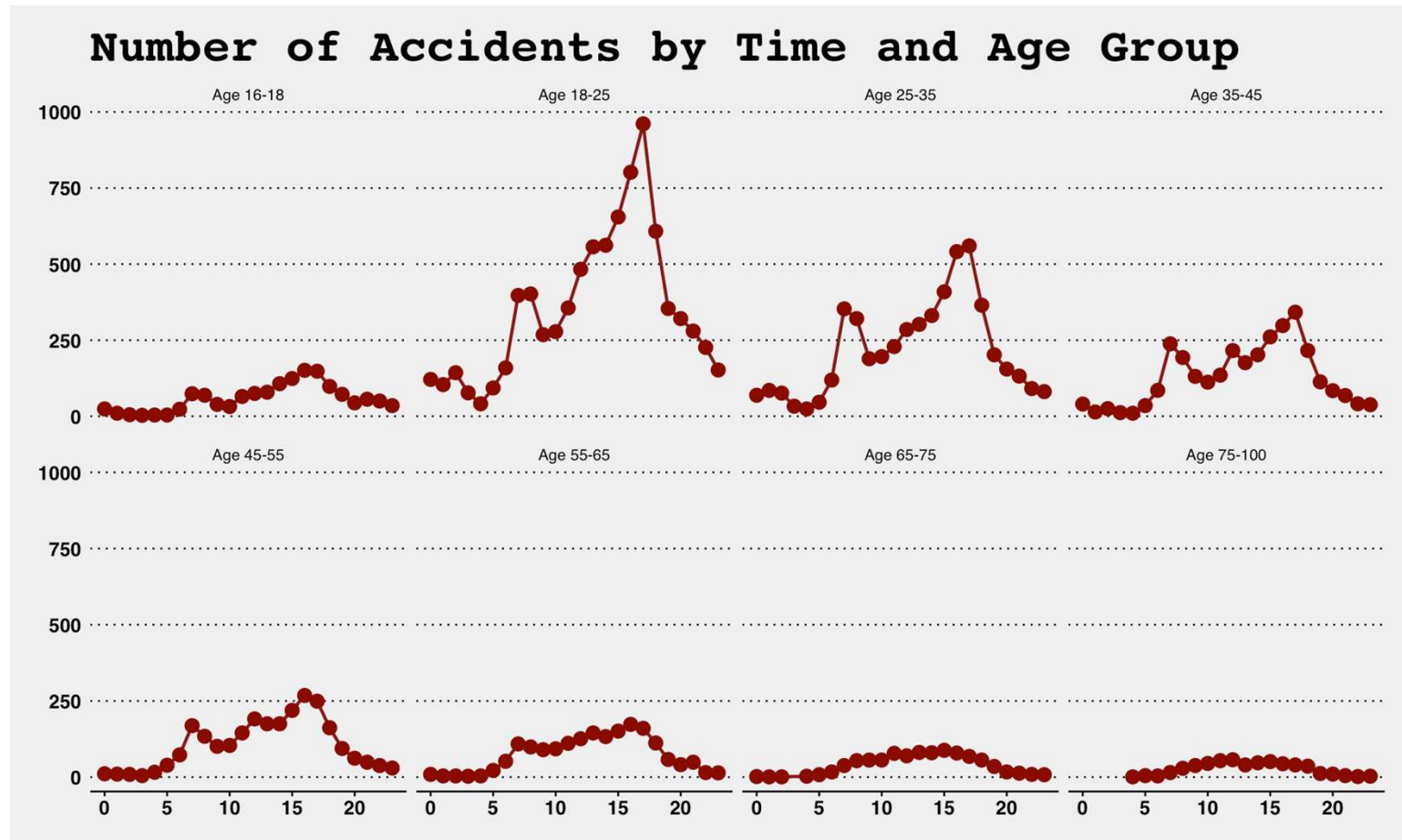
Visualizations for date might be contingent upon the type. Heatmap makes sense for time.



Calendar Heat Map of Birth Date of MLB Players



Days of week and hours of the day



thank you

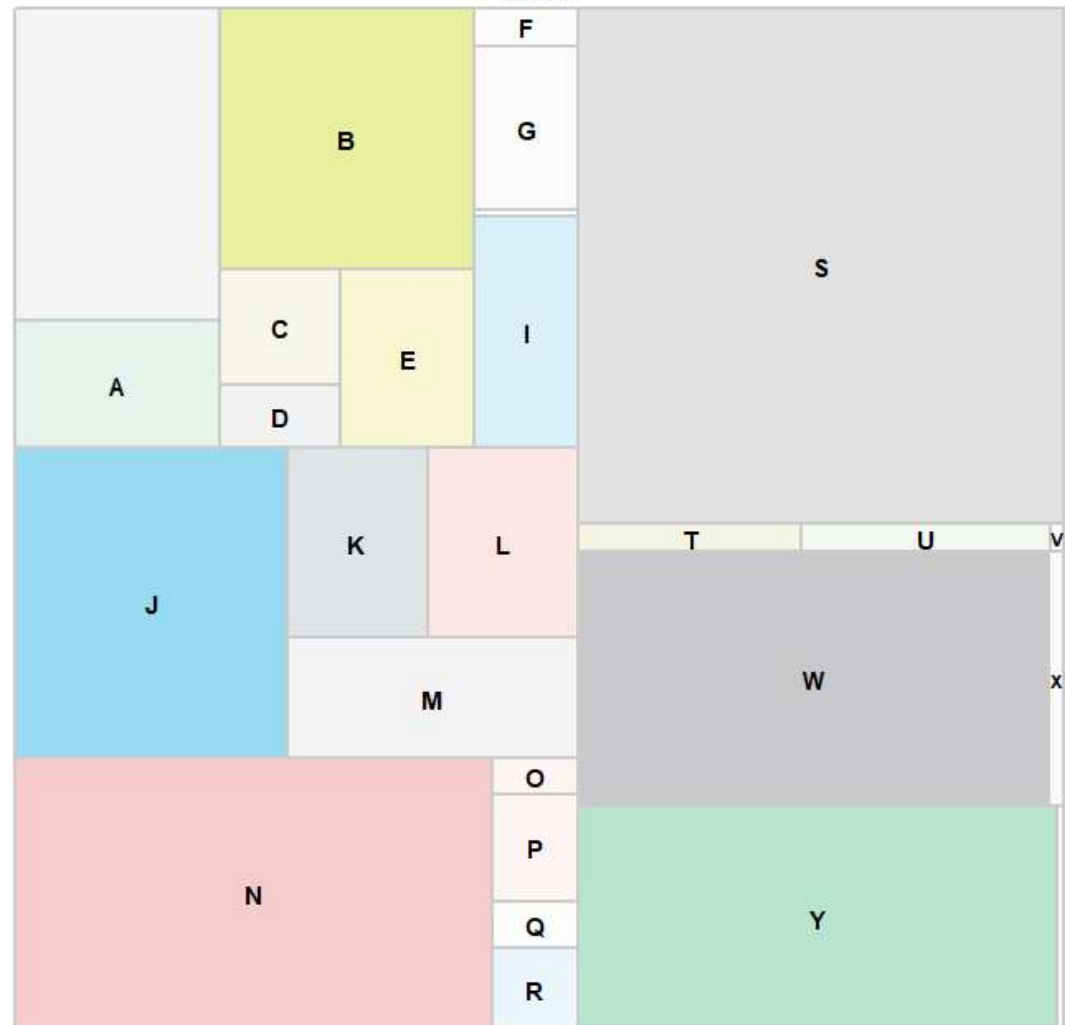
Formatting Experiments

Here are all of the pirate palettes

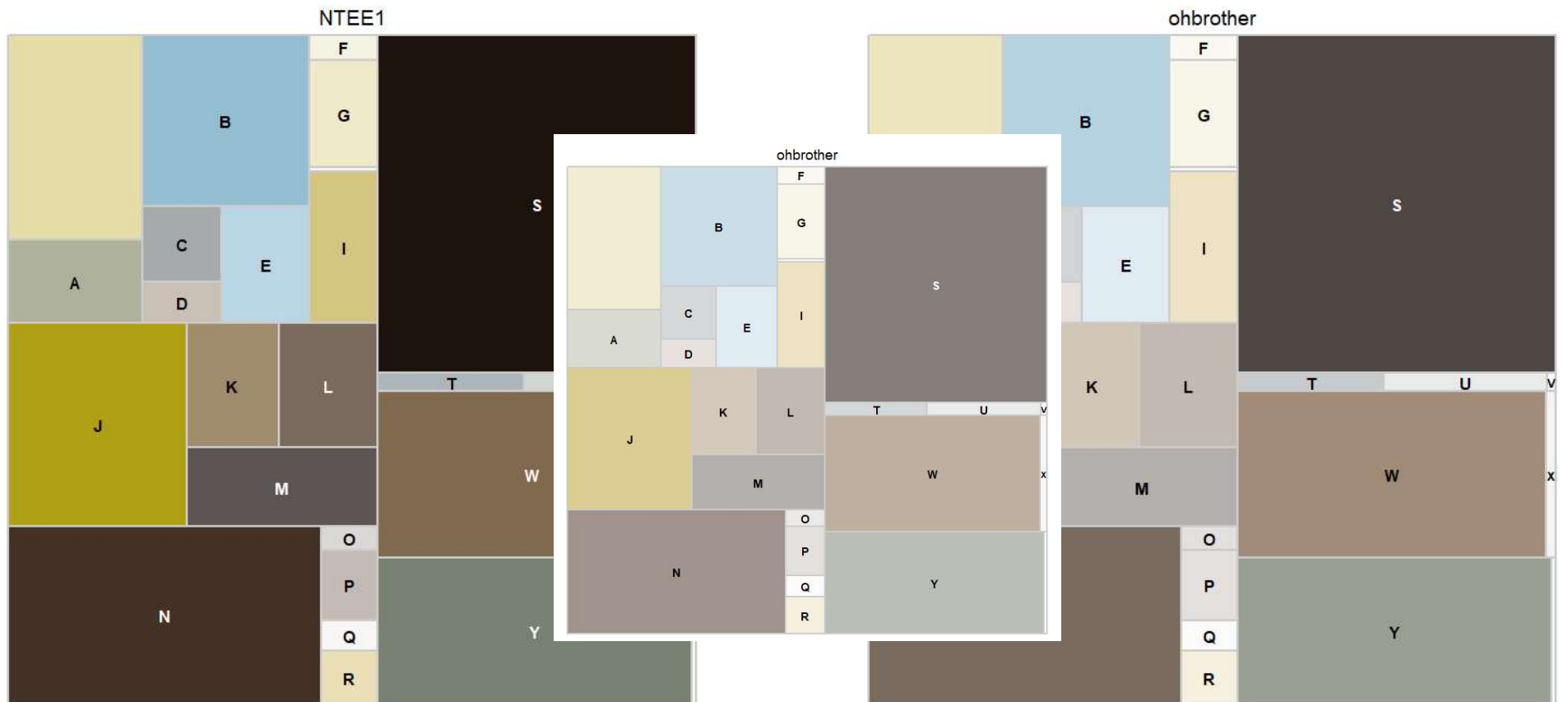
Transparency is set to 0



ipod

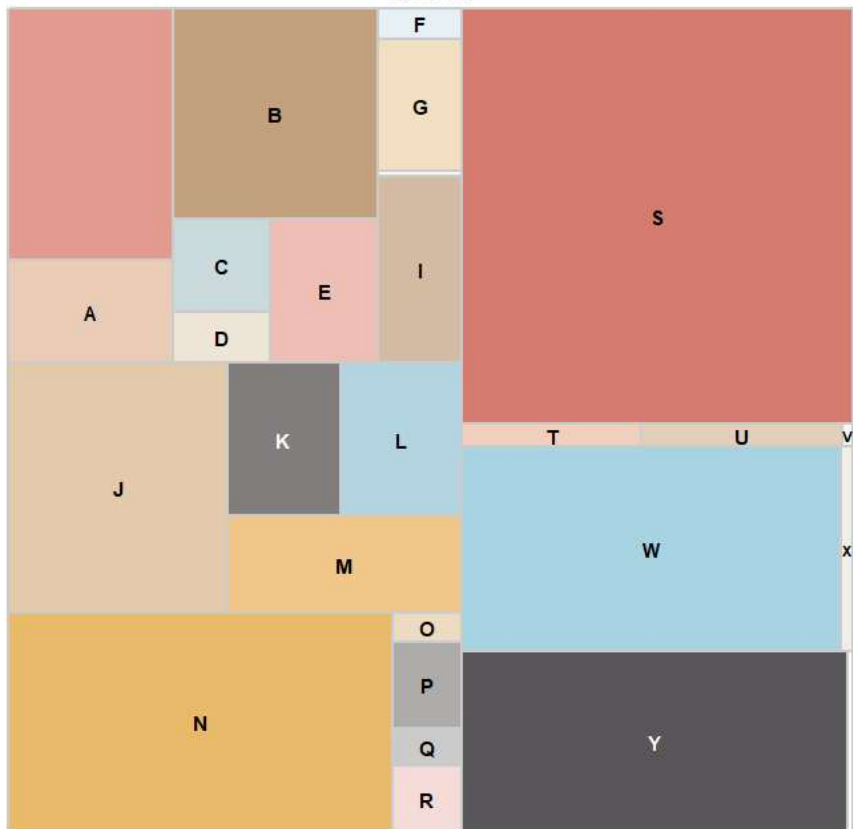


ohbrother

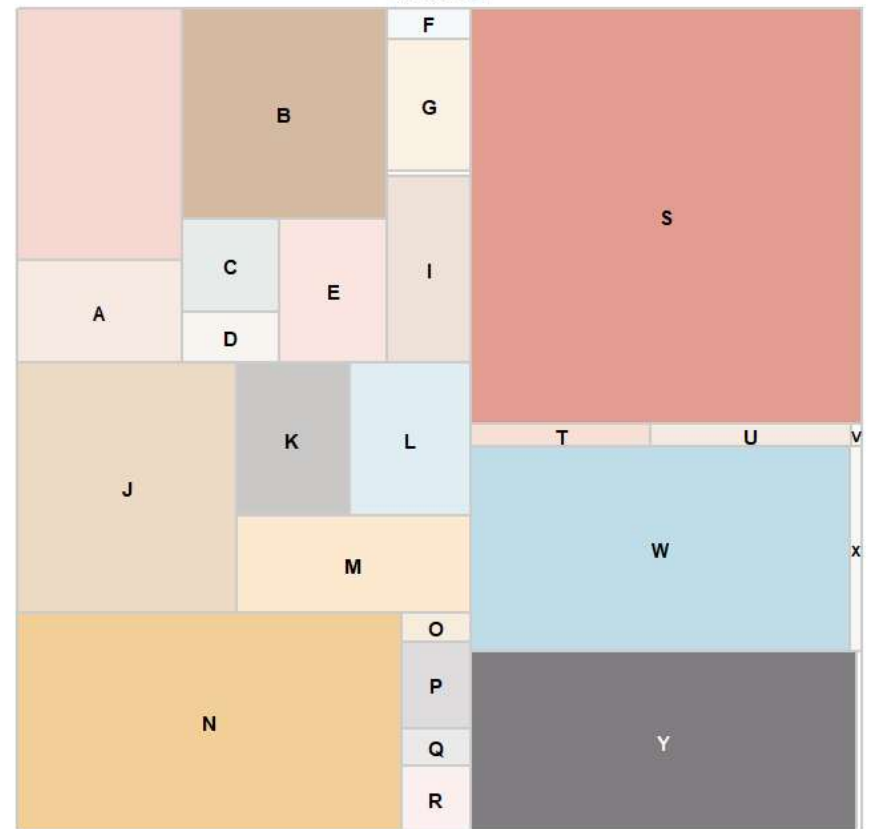


decision

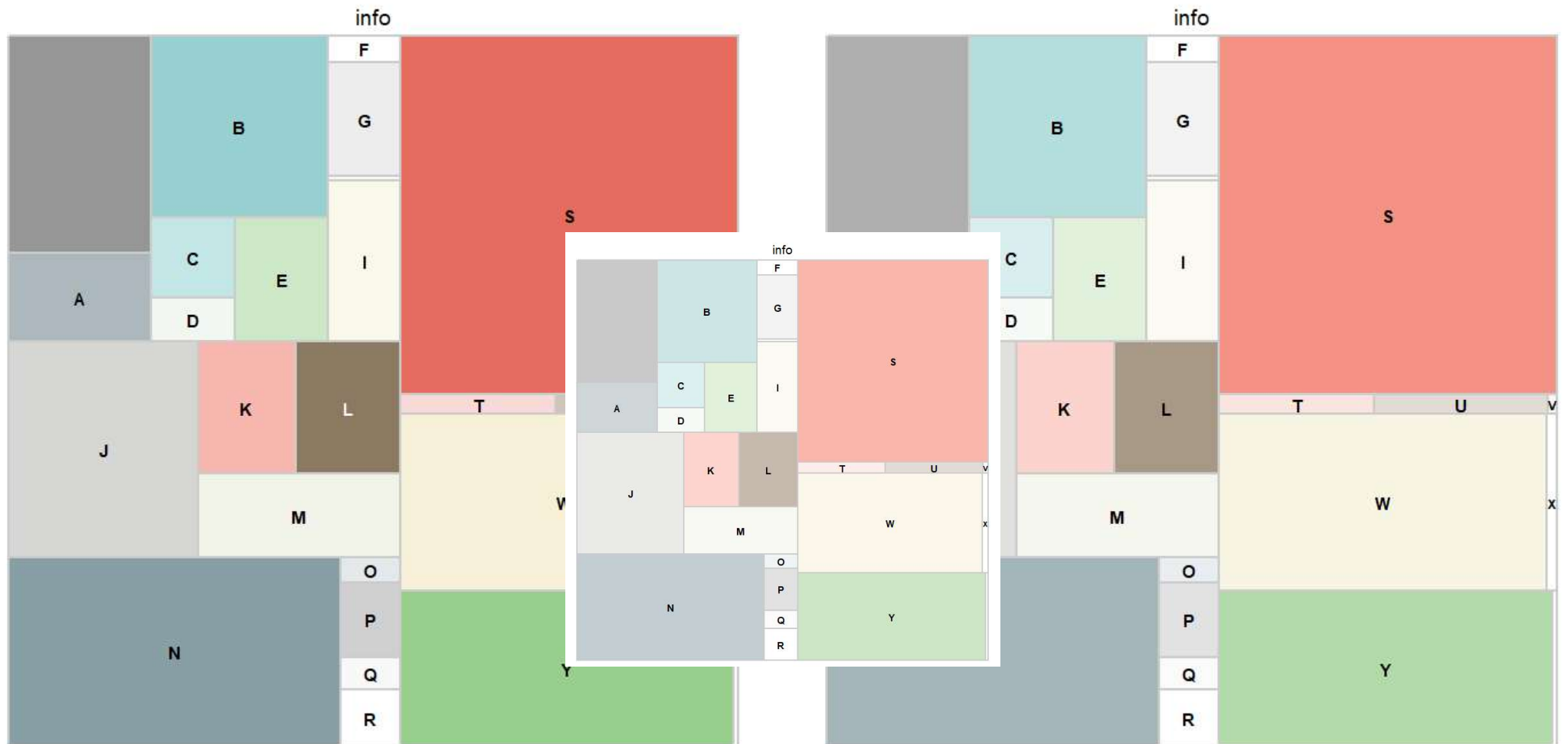
NTEE1

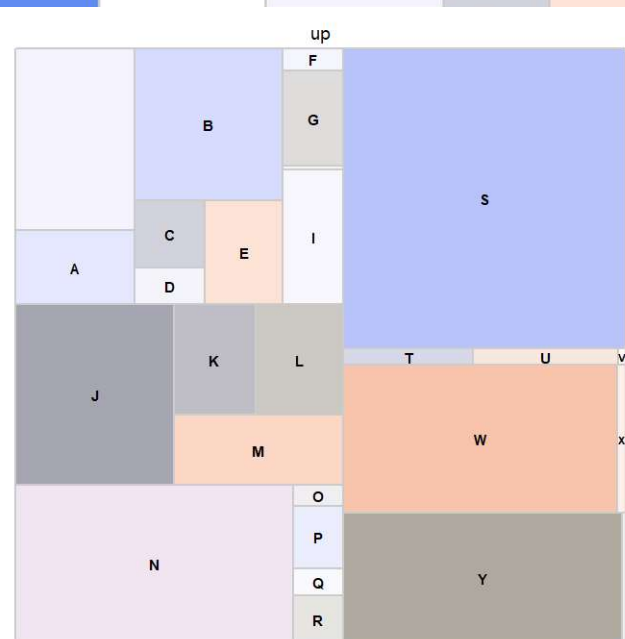
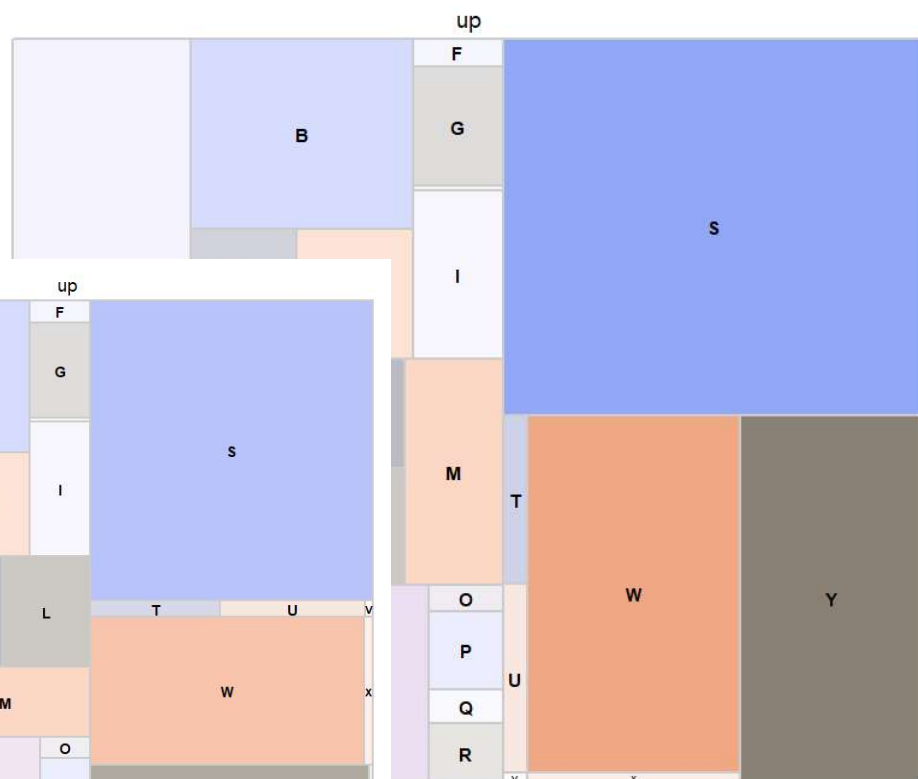
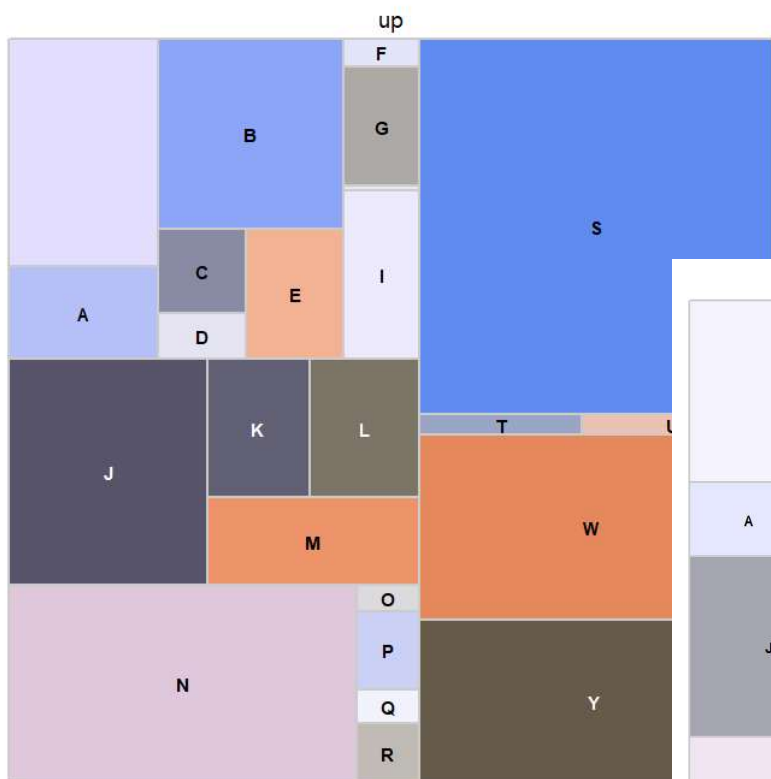


decision

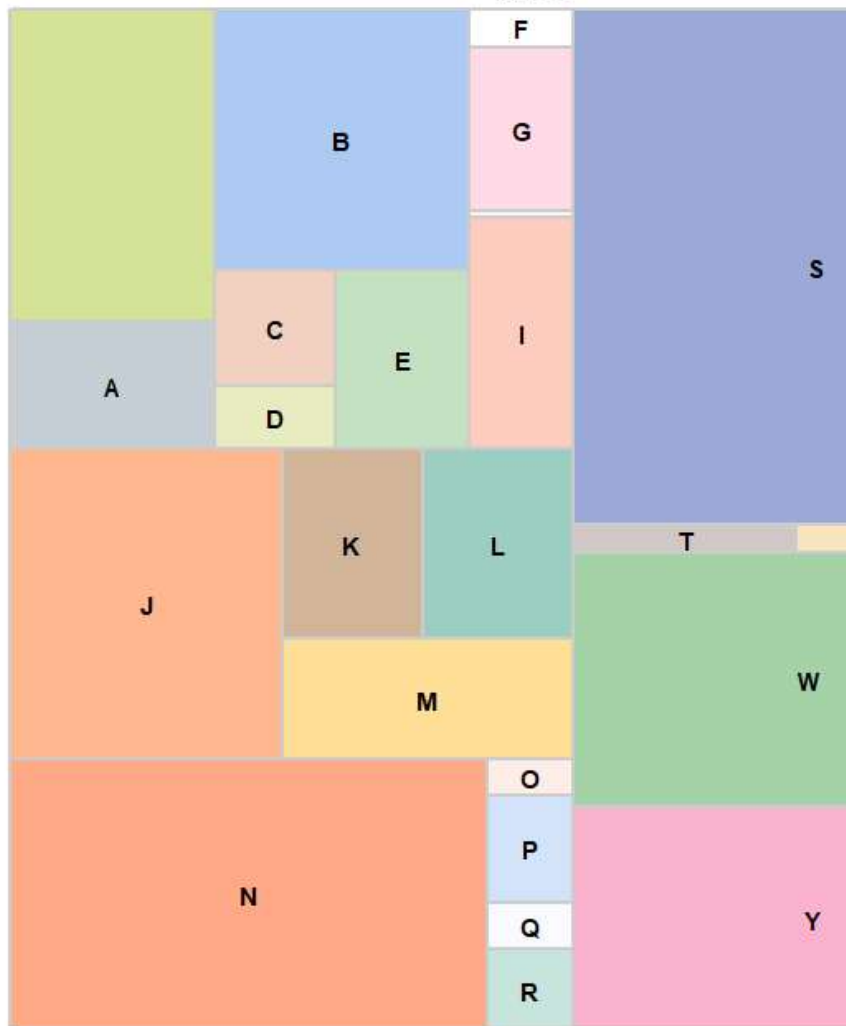


info





basel



usualsuspects

